Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 (Generated practice paper - Set 40)

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| TOTAL | |

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**Answer all questions in the spaces provided.**

**0 1 . 1** Explain why **proxy data** are needed to study the early conditions on Earth.

**[1 mark]**

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**0 1 . 2** Complete Table 1 to show what each analytical technique can be used to estimate.

**[4 marks]**

**Table 1**

| Technique | What it estimates | |
|---|---|---|
| Ratio of carbon-12 to carbon-14 in a sample | | |
| Ratio of oxygen-18 to oxygen-16 | | |
| Analysis of air bubbles trapped in ice cores | | |
| Ratio of magnesium to calcium in calcite deposits | | |

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**0 1 . 3** Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

**[2 marks]**

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**0 1 . 4** Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

**[3 marks]**

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**0 2** The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where $S_1$ and $S_2$ are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

**Table 1**

| Term | Biomass / thousand tonnes |
|---|---|
| $S_1$ | 600 |
| A | 90 |
| G | 150 |
| C | 220 |
| M | 80 |

**0 2 . 1** Calculate $S_2$ and state whether the stock is being fished sustainably.

**[2 marks]**

**Answer** ☐ thousand tonnes

**0 2 . 2** Calculate the maximum catch that would have kept the stock constant.

**[1 mark]**

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**Answer** [ ] thousand tonnes

**0 2 . 3** Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

**[3 marks]**

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**0 2 . 4** Describe how each of the following measures obtained from commercial catches can indicate overfishing.

**[4 marks]**

**Catch per unit fishing effort** .....

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**Mean fish size and mean age** .....

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**0 3 . 1** Complete Table 1 to compare natural systems with typical linear human systems.

**[4 marks]**

**Table 1**

| Feature | Natural systems | Linear human systems |
|---|---|---|
| Energy source | | |
| Use of materials | | |
| Nature of wastes | | |
| Temperature of processes | | |

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**0 3 . 2** Explain why natural systems can be sustained using a relatively small number of elements.

**[2 marks]**

**0 3 . 3** Explain why the human use of minerals makes them difficult to recover and re-use.

**[2 marks]**

**0 3 . 4** Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

**[2 marks]**

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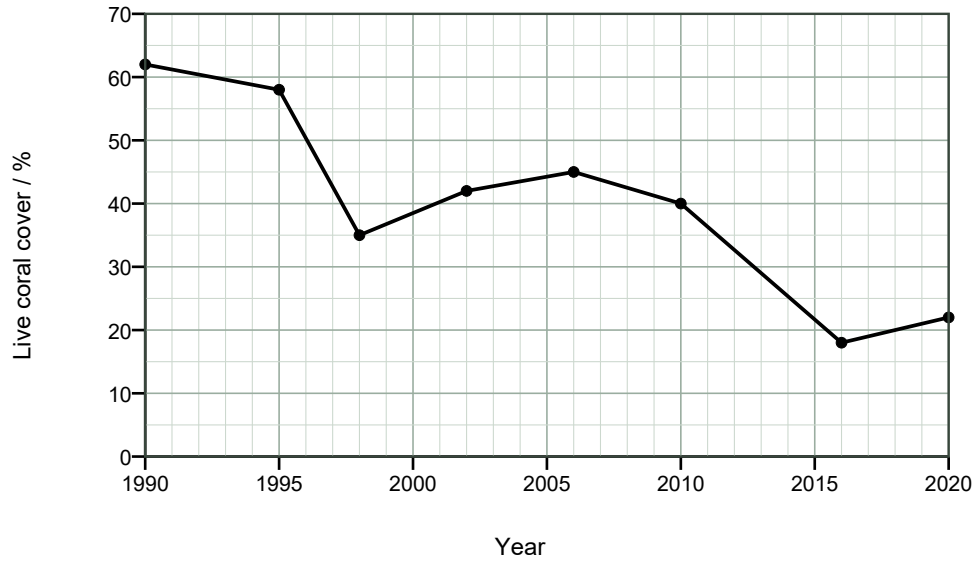
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**0 4** Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

**Figure 1**




**0 4 . 1** Describe the changes in coral cover shown in Figure 1.

**[2 marks]**

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**0 4 . 2** Explain what coral bleaching is and give **three** causes.

**[4 marks]**

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**0 4 . 3** Explain how the growing resort could threaten the reef other than through bleaching.

**[3 marks]**

**0 4 . 4** Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

**[1 mark]**

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**0 5** Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

**Table 1**

| Treatment | Mean dry mass / g | Standard deviation / g |
|---|---|---|
| Organic manure | 14.2 | 3.9 |
| Inorganic NPK fertiliser | 16.1 | 4.1 |

**0 5 . 1** Explain what the standard deviation values tell you about the data.

**[2 marks]**

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**0 5 . 2** Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

**[2 marks]**

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**0 5 . 3** The test gave a probability p = 0.30. State what this means and what the student should conclude.

**[2 marks]**

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**0 5 . 4** Explain why a statistically significant correlation does not prove a causal relationship.

**[2 marks]**

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**0 5 . 5** Suggest how the student could increase the chance of detecting a real difference if one exists.

**[2 marks]**

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**0 6 . 1** Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

**[1 mark]**

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

**0 6 . 2** Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

**[2 marks]**

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**0 6 . 3** Explain how DNA databases could help to control the illegal trade in elephant ivory.

**[2 marks]**

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**0 7** Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

**Table 1**

| Method | Mean yield / t $ha^{-1}$ | Pesticide applications per year | Non-target insect species recorded in field margins |
|---|---|---|---|
| Chemical insecticide only | 18.5 | 6 | 8 |
| Crop rotation + beetle banks + hedgerows | 16.2 | 0 | 31 |
| Integrated control | 18.1 | 1-2 | 26 |

**0 7 . 1** Explain how crop rotation and beetle banks reduce pest damage.

**[4 marks]**

**0 7 . 2** Use Table 1 to evaluate integrated control compared with the other two methods.

**[3 marks]**

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**0 7 . 3** Describe how the sterile male technique controls insect pests and explain why it is species-specific.

**[3 marks]**

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**0 8** The Isle of Eigg (area 31 km$^2$, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

**Table 1**

| Source | Capacity / kW |
|---|---|
| Hydroelectric (3 generators) | 110 |
| Solar PV | 50 |
| Wind (4 turbines) | 24 |
| Diesel generators (back-up) | 160 |

**0 8 . 1** Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

**[3 marks]**

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**0 8 . 2** Describe **three** other ways in which the Eigg community has increased its sustainability.

**[3 marks]**

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**0 8 . 3** Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

**[2 marks]**

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**0 8 . 4** Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

**[2 marks]**

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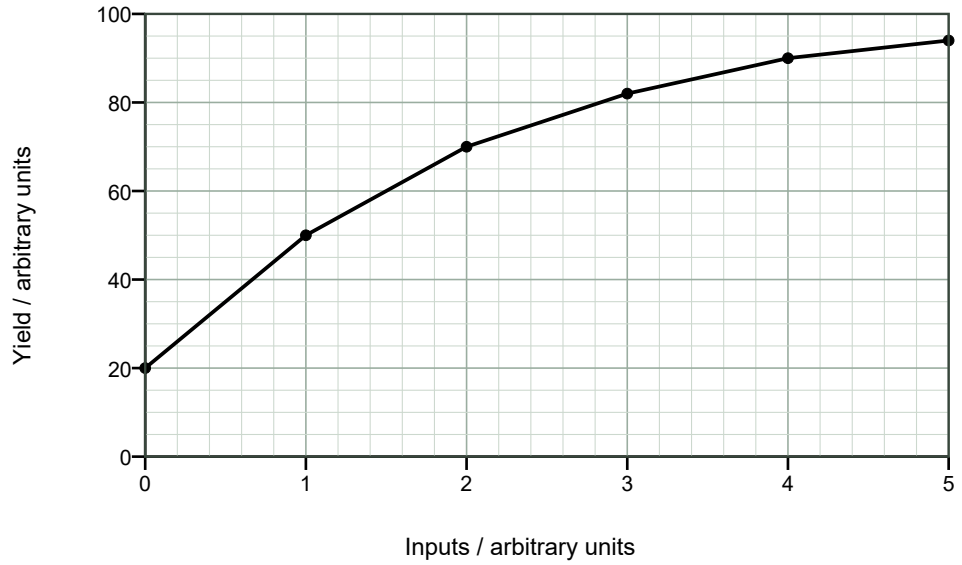
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**0 9** Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

**Figure 1**




**0 9 . 1** Describe the relationship shown in Figure 1.

**[2 marks]**

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**0 9 . 2** Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

**[2 marks]**

**Answer** _____ units

**0 9 . 3** Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

**[2 marks]**

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**0 9 . 4** Suggest why inputs are nevertheless often used unevenly across the world.

**[2 marks]**

**0 9 . 5** Explain why intensive agriculture may be very productive but not very efficient.

**[2 marks]**

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**1 0 . 1** Discuss the reasons why the conservation of biodiversity is important for human society.

**[9 marks]**

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**1 0 . 2** State what is meant by **biodiversity**.

**[1 mark]**

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**Write an essay on one of the following topics.**

**11 . 1** Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

**[25 marks]**

**OR**

**11 . 2** Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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**END OF QUESTIONS**